Library Management System (Java) Project Report

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1. Introduction

Libraries are essential for academic and personal growth. However, traditional manual systems for managing books are inefficient and prone to errors.

This project presents a Library Management System developed in Java to automate book management tasks.

2. Problem Statement

Manual systems cause delays, errors, and poor record maintenance.

This project aims to provide a simple digital solution to manage books efficiently.

3. Objectives

• Automate library operations

• Manage book records efficiently

• Apply Java OOP concepts

• Provide a simple user interface

4. Technologies Used

Programming Language: Java

Concepts: OOP, ArrayList, Loops, Conditions

5. System Design

Book Class: Stores book details

Library Class: Manages collection of books

Main Class: Provides menu interface

6. Features

• Add new books

• View all books

• Issue books

• Return books

• Track availability

7. Working

1. User runs the program

2. Menu is displayed

3. User selects operation

4. System performs action

8. Advantages

• Reduces manual work

• Fast and efficient

• Easy to use

9. Limitations

• No database storage

• CLI based interface

10. Future Enhancements

• GUI using Java Swing

• Database integration

• Login system

11. Conclusion

The system successfully automates library tasks and demonstrates Java programming concepts.